THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 1: **A Square and a Cube** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a perfect square? 1

(a) 300 **(b)** 230 **(c)** 225 **(d)** 250

2. The value of 17^2 is: 1

(a) 289 **(b)** 272 **(c)** 306 **(d)** 34

3. $\sqrt{324}$ is equal to: 1

(a) 20 **(b)** 17 **(c)** 19 **(d)** 18

4. $\sqrt{1764}$ is equal to: 1

(a) 44 **(b)** 43 **(c)** 42 **(d)** 41

5. The sum of the first 20 odd natural numbers is: 1

(a) 420 **(b)** 399 **(c)** 400 **(d)** 40

6. The value of 6^3 is: 1

(a) 222 **(b)** 210 **(c)** 108 **(d)** 216

7. $\sqrt[3]{343}$ is: 1

(a) 7 **(b)** 6 **(c)** 8 **(d)** 9

8. $\sqrt[3]{9261}$ is: 1

(a) 23 **(b)** 22 **(c)** 20 **(d)** 21

9. The ones digit of a perfect square can never be: 1

(a) 7 **(b)** 4 **(c)** 1 **(d)** 9

10. The cube of a negative number is: 1

(a) always even **(b)** zero **(c)** positive **(d)** negative

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write the sum of the first 9 odd natural numbers without adding. 2

12. Write the sum of the first 12 odd natural numbers without adding. 2

13. What is a perfect square? Give two examples. 2

14. What is a perfect cube? Give two examples. 2

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SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find $\sqrt{4624}$ using prime factorisation. 3

16. Find $\sqrt{7396}$ using prime factorisation. 3

17. Find the smallest number by which 180 must be multiplied to make it a perfect square. Also find the square root of the product. 3

OR

Find the smallest number by which 500 must be divided to make it a perfect cube. 3

18. Express 64 as the sum of consecutive odd numbers starting with 1. 3

OR

Find a Pythagorean triplet whose smallest member is 9. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Write the Pythagorean triplets obtained from $m = 5$, 7 and 9 using $\left(m, \frac{m^2-1}{2}, \frac{m^2+1}{2}\right)$. 5
(ii) Verify one of them.

OR

(i) Prove that n^3 can be written as the sum of n consecutive odd numbers; show it for $n = 4$. 5
(ii) Write 5^3 in this way.

20. **Case study:** Hardy visited Ramanujan in hospital and mentioned that his taxi number was 1729. Ramanujan said it is a very interesting number. 5
(i) Why is 1729 interesting?
(ii) Express it as a sum of two cubes in both ways.
(iii) Verify each sum.

OR

(i) Find the square root of $\frac{441}{784}$. 5

(ii) Find $\sqrt[3]{\frac{343}{1728}}$.

END OF QUESTION PAPER — Total: 40 Marks

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